W-F-408D
30 March 1984
SUPERSEDING
W-F-408C
August 16, 1972

FEDERAL SPECIFICATION

# FITTINGS FOR CONDUIT, METAL, RIGID (THICK-WALL AND THIN-WALL (EMT) TYPE)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers raintight, concrete tight, and miscellaneous fittings for rigid metal conduit and electrical metallic tubing. This specification does not cover fittings designed specifically to meet the requirements of theNational Electrical Code for use in hazardous locations or cast metal outlet bodies and floor boxes.

1.2 Classification.

1.2.1 Types, classes, kinds, styles, sizes, and materials. Fittings furnished under this specification shall be of the following types, classes, kinds, styles, materials, and sizes, as specified (see 6.2).

NOTE: Classes, kinds, styles, and materials of the Type I and Type II fittings are identical except for Type II, kind G.

Type I - Raintight (see 6.4).

Class 1 - Fittings for thick-wall conduit
- Kind A - Coupling, electrical conduit threadless
- Kind B - Box connector, electrical, straight, threadless
  - Style 1 - Unisulated
  - Style 2 - With insulated throat
- Kind X - Box connector, (hub), electrical, straight, threaded
  - Style 1 - Uninsulated
  - Style 2 - With insulated throat
- Kind C - Box connector, electrical, 90°, threaded
  - Style 1 - Uninsulated
  - Style 2 - With insulated throat
- Kind D - Box connector, electrical, 45°, threaded
  - Style 1 - Uninsulated
  - Style 2 - With insulated throat
- Kind E - Box connector, electrical, 90°, threadless
  - Style 1 - Uninsulated
  - Style 2 - With insulated throat



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Kind F - Box connector, electrical, 45°, threadless
Style 1 - Uninsulated
Style 2 - With insulated throat
Kind H - Adapter, electrical conduit
Style 3 - Enlarger
Style 4 - Reducer
Kind J - Offset connector, electrical conduit, threaded
Style 5 - External to internal
Style 6 - External to external

Class 2 - Threadless fittings for thin-wall conduit (EMT)
Kind K - Coupling, electrical conduit
Kind L - Box connector, electrical, straight, with single locknut
Style 1 - Uninsulated
Style 2 - With insulated throat
Kind M - Box connector, electrical, 90°
Style 1 - Uninsulated
Style 2 - With insulated throat
Kind N - Box connector, electrical, 45°
Style 1 - Uninsulated
Style 2 - With insulated throat
Kind Y - Offset connector, electrical conduit

Type II - Concrete tight (see 6.4)

Class 1 - Fittings for thick-wall conduit
Kind G - Union, electrical conduit

Type III - Miscellaneous fittings

Class 1 - Fittings for thick-wall conduit
Kind P - Locknut, electrical conduit
Kind Q - Bushing, electrical conduit, metal
Style 1 - Uninsulated
Style 2 - With insulated throat
Style 7 - Capped
Style 8 - Grounding
Style 9 - Grounding, insulated
Kind R - Bushing, electrical conduit, nonmetallic
Kind S - Bushing, electrical conductor, nonmetallic
Kind T - Chase nipple, conduit, metal
Style 1 - Uninsulated
Style 2 - With insulated throat

Class 2 - Threadless fittings for thin-wall conduit (EMT)
Kind U - Bushing, electrical conduit, metal
Style 1 - Uninsulated
Style 2 - With insulated throat

Sizes

0.50 - 1/2 inch
0.75 - 3/4 inch
1.00 - 1 inch
1.25 - 1-1/4 inch



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1.50 - 1-1/2 inch
2.00 - 2 inch
2.50 - 2-1/2 inch
3.00 - 3 inch
3.50 - 3-1/2 inch
4.00 - 4 inch

Materials

Steel
Malleable Iron
Cast Iron
Aluminum alloy
Zinc alloy

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards

FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specification, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Decription. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item decriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC, Atlanta, GA Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.



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(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of the invitation for bids or request for proposal shall apply.

American Society Testing and Materials (ASTM) Standard

B86 - Standard Specification for Zinc-Alloy Die Castings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

Underwriters Laboratories, Inc. (UL)

UL 467 - Standard for Grounding and Bonding Equipment
UL 514B - Fittings for Conduit and Outlet Boxes

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of procedures. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

3. REQUIREMENTS

3.1 Codes and standards. The fittings shall conform to the requirements of UL 514B and UL 467 as applicable.

3.1.1 Compliance Prior to approval of the first shipment, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the fittings he proposed to furnish under this specification meets the requirements of UL 514B and UL 467 as applicable.

3.1.2 Underwriters Laboratories, Inc. Acceptable evidence of meeting the requirements of UL 514B and UL 467 shall be the UL certification symbol or label, listing in the Listing File, or a certified test report from a recognized independent testing laboratory indicating the fitting has been tested and conforms to UL 514B and Ul 467. Such evidence must be acceptable to the contracting officer.

3.2 Material. Materials used shall be free from defects which would adversely affect the performance of maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the



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same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the mazimum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.2.1 Metallic materials. Fittings and component parts of fittings shall be made of steel, malleable iron, cast iron, or not-ferrous metals as specified. Fittings made from die-cast Zinc material shall conform to ASTM B86.

3.2.2 Insulating material. Insulating material used for fittings and component parts of fittings shall be suitable for use in concrete, and shall conform to the requirements of UL514B.

3.3 Standard commercial product. The conduit fittings shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibitied by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the conduit fittings being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Electrical continuity. A fitting shall join with other parts of the raceway system in such a manner as to provide continuous electrical conductivity when tested for Electrical Continuity in accordance with 4.6.

3.6 Grounding and bonding. Fittings and accessories furnished under this specification shall meet all the applicable requirements stated in UL 467.

3.7 Identification marking. Each fitting and accessory shall be marked in accordance with the requirements set forth in UL 467 and UL 514B.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3)
b. Preparation for delivery inspection (see 4.7)

4.3 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and tests of 4.6. This inspection shall be performed on the samples selected in accordance with 4.4.

4.4 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All fittings offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.1 Sampling for examination. Examination shall be based on inspection level II and an acceptable quality level (AQL) of 1.5 percent defective.

4.4.2 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 1.0 percent defective.

4.5 Examination. Each fitting shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examintations and dimensional measurements. Noncompliance with any specified requirement shall constitute a defect.

4.5.1 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.1.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested in accordance with UL 514B, as applicable for the type specified.

4.7 Preparation of delivery inspection. The preservation, packaging, and packing of the item shall be inspected to verify compliance to the quality conformances inspection requirements of MIL-E-17555.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing. Preservation and packing shall be in accordance with the requirements of MIL-E-17555 with the level of preservation and the level of packing as specified (see 6.2).



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5.2. Marking.

5.2.1 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Raintight fittings are intended for outdoor use. Concrete tight fittings are intended for embedment in fresh concrete. A raintight fitting is also considered to be concrete tight. Raintight and concrete tight fittings are suitable for general use.

6.2 Ordering data requirements. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification
b. Type, class, kind, style, size, and material of fittings required (see 1.2.1)
c. Level of preservation and Level of packing required (see 5.1)

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-1004.9(n) of the Defense Acquisition Regulations (DAR), the data Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Definitions.

Raintight fitting. A raintight tight fitting is one so designed that, when assembled in the intended manner and exposed to a beating rain, it will not permit the entrance of water into the interior of the fitting.

Concrete tight fitting. A concrete tight fitting is one so designed that, when assembled in the intended manner and embedded in freshly mixed concrete, it will not permit the entrance of cement into the interior of the fitting.

6.5 Cross reference.

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|---|---|
| Paragraph 1.2.1 | Paragraph 1.2.1 |
| Kind BB | Kind X |
| Kind O | Kind Y |



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MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 85

Review Activities

Army - ME
AF-99
DLA-GS

User Activities:

Army - CE
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS, PBS

PREPARING ACTIVITY:

Navy - YD

DOD project 5975-0558

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies an other documents referenced herein.



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